Article

Agentic Lightweight Consensus for Resilient Monitoring and Control of Modern Data Centers in Smart Cities

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Highlights

What are the main findings?

- Across 1,890 replay scenarios, no estimator dominates globally; persistent FPR–OWA+CRP provides the strongest anomaly diagnostic, while an aligned zone-dynamic variant reduces non-convergence but slightly worsens MAE relative to social-only reliability.
- The tested command gate blocks every observed unsafe or process-invalid proposal. An illustrative periodic closed-loop pilot reports tracking, SLA-exposure, recovery, and actuator-effort metrics and reveals delayed recovery under common-mode sensor bias.

What are the implications of the main findings?

- Sensing consensus, persistent diagnostics, supervisory proposal, and actuation admission should remain separate, traceable roles.
- The evidence supports software-level runtime validation and bounded simulation claims; physical safety, production authentication, energy consumption, and a causal LLM advantage remain unvalidated.

Abstract

Data-center monitoring for smart-city services must fuse conflicting sensor reports and keep AI-proposed commands outside the safety-critical path. We present a two-path architecture in which a deterministic fast path combines reliability-based fuzzy-preference OWA (FPR–OWA), a persistent consensus-reaching process (CRP), and an actuation verifier; an optional reactive or language-model supervisor can only propose typed actions. Across 1,890 replay scenarios and seven attack families, no estimator dominates: averaging has the lowest global error and sensor redundancy explains most accuracy variation, while FPR–OWA+CRP provides the strongest diagnostic signal. A separate action-dependent simulation adds standard control metrics and exposes an adverse common-mode-bias boundary: the measurement-dependent controllers recover much later than the oracle. The verifier blocks every unsafe or process-invalid command observed in the historical safety set and deterministic fault-injection matrix, an empirical result rather than a formal system guarantee. A true-CRP benchmark remains below 0.30 ms median latency at $n_z = 100$ in the tested environment. Historical reactive and LLM results are retained only as descriptive, non-paired evidence; no causal LLM advantage, physical-hardware validation, production authentication, or energy saving is claimed.

Keywords: smart cities; urban critical infrastructure; agentic AI; multi-agent systems; data-center cyber–physical systems; soft consensus; fuzzy preference relations; runtime verification

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1. Introduction

Data centers sustain smart-city services across cloud and edge infrastructure, but their control loops depend on thermal, power, and workload telemetry that may be noisy, missing, stale, or deliberately deceptive [1,2,7]. The concrete problem addressed here is therefore twofold: at every control tick, a zone must infer a trustworthy state from conflicting reporters; before actuation, it must prevent an autonomous supervisor from turning uncertain evidence or malformed output into an unsafe command.

Existing methods solve only parts of this problem. Hierarchical data-center monitoring and multi-agent management distribute collection and resource decisions but do not model graded trust under deceptive sensing [8–11]. Robust estimation and resilient agreement tolerate outliers yet usually do not emit the persistent, auditable consensus diagnostics needed by a supervisory control path [42–44]. Soft-consensus models provide graded agreement and feedback but were developed for human group decision making rather than per-tick cyber–physical telemetry [31,36]. LLM agent frameworks add flexible tool use, while runtime-assurance and policy-enforcement systems separate proposal from execution; neither removes the need to validate the state on which a command is based [37,45,46].

The resulting gap is a tested composition of (i) trust-graded sensor fusion, (ii) persistent consensus diagnostics, and (iii) deterministic proposal-to-actuator admission under one traceable runtime contract. We address this gap with a zone-level architecture for data-center thermal monitoring. The contribution is the integration and evaluation of these mechanisms; it is not a claim of a universally superior estimator, a production security boundary, or a hardware-validated controller.

The paper addresses five questions:

- **RQ1, estimation:** how do candidate aggregators behave under seven deception families, and which factors drive error?
- **RQ2, diagnostics:** what diagnostic value is added by persistent CRP monitoring, including self-history and zone-dynamic alternatives?
- **RQ3, command admission:** which unsafe or process-invalid commands are blocked in the tested safety episodes and fault-injection matrix?
- **RQ4, supervisory role:** what can be concluded from the available reactive and LLM records, given that their historical execution paths are not a paired experiment?
- **RQ5, operational cost and control:** what latency, memory, tracking, SLA-exposure, recovery, and actuator-effort behavior is observed in the bounded software evaluation?

The contributions are:

1. a two-path architecture with non-overlapping sensing, consensus, CRP-monitoring, supervisory-proposal, command-admission, fallback, and audit roles;
2. an FPR–OWA operator with persistent CRP feedback, together with a formal proof that logistic dominance preserves the reliability order and a frozen-trajectory temporal ablation;
3. a runtime contract that binds immutable snapshots to typed proposals and admits actuation only through a deterministic, one-shot authorization, with explicit fallback and traceability; and
4. a reproducible evidence package comprising 1,890 replay scenarios, deterministic fault injection, an illustrative action-dependent thermal pilot and hot-start boundary, and a true-CRP latency/memory benchmark.

The evaluation deliberately separates historical replay, historical non-paired supervisor records, deterministic runtime tests, and the new periodic closed-loop pilot. This separation prevents causal claims from crossing incompatible execution paths. Section 2 critically positions the work; Sections 3–5 specify the architecture and algorithms; Sections 6–7 report the evidence; and Sections 8–10 state its implications and limits.

2. Related Work

This work intersects four research lines: consensus for blockchain-IoT integration, hierarchical monitoring and multi-agent management of data centers, soft consensus in fuzzy group decision making, and LLM-based agentic systems. Although these areas have largely developed in parallel, this selective review focuses on the elements needed to clarify the research gap addressed in this paper.

2.1. Consensus in Blockchain–IoT Integration

Blockchain-IoT integration has produced a range of consensus mechanisms for constrained devices, including authority-based validation, edge-hosted ledgers, hybrid on-chain/off-chain patterns, and energy-recycling schemes [17–23]. Architecturally similar, Ramachandran's three-tier hierarchical framework maps device classes to specific consensus roles, using district PBFT [24] under city-wide proof of authority [16]. Across these methods, the common object of agreement is the ledger, and the purpose is to guarantee transaction ordering, immutability, and quorum bounds. The target problem considered here differs: it concerns estimation of the physical state from graded, partially deceptive reports at each control round. Ledger consensus is therefore treated as a complement rather than a baseline (Section 2.5). Outside the ledger domain, Zhang et al. design approximate consensus for small devices in lossy networks [15]. We follow a similarly lightweight direction while explicitly modeling reporter trust, which that framework does not provide.

2.2. Data-center Monitoring, Multi-Agent Management, and Safe Control

Hierarchical sensing is well established in data centers. Jiang et al. partition an internet data center into logical grids, with edge nodes preprocessing environmental and host telemetry before escalation [8]; Yan et al. combine edge computing, wireless sensing, and IPMI-derived measurements for environmental monitoring [9]. For infrastructure management, the Unity system decomposes power and performance management into interacting agents [10], consistent with the broader autonomic-computing paradigm of decentralized, self-managing elements [11]. Multi-layer IoT/edge coordination based on compact summaries rather than raw-state collection is likewise well documented [40,41]. Production-trace characterization confirms the coupled workload, energy, and temperature dynamics that these hierarchies must track [7].

For actuation, deep reinforcement learning has substantially reduced cooling energy, but unconstrained exploration can violate thermal safety. Physics-guided safe RL mitigates this risk by rectifying recommended actions against a learned thermal model [12], whereas event-driven formulations reduce unnecessary control operations [13]. These findings motivate two design commitments in our architecture: proposed actions must be validated against explicit safety constraints before actuation, and the state supplied to the controller must itself be validated against noise and deception [14].

Beyond the facility, data centers are increasingly studied as urban infrastructure. Edge data centers are deployed across cities according to mobility and demand patterns [2]; smart-city service stacks rely on the resulting edge–cloud continuum [1]; facility thermal behavior couples with urban energy systems through waste-heat reuse in district heating [6]; and urban governance research frames data centers as the material backbone of smart cities, whose footprint cities now actively regulate [5]. The monitoring and control layer examined in this paper operates within the facility but provides a reliability foundation for these urban integrations: district-energy coupling, urban edge offloading, and city-level orchestration all presuppose trustworthy reporting of the facility state.

Two additional bodies of work inform this framing. The first concerns resilient estimation and aggregation. Secure state estimation under false-data injection [42,43], robust distributed agreement such as the W-MSR mean-subsequence-reduced rule [44], and classical robust statistics, including M-estimators and median/MAD methods, provide the baseline theory for our fuzzy-consensus operator. We position FPR–OWA as a trust-graded variant within this space rather than as a categorically new estimator. The second concerns runtime assurance and safe LLM control. Our deterministic verifier acts as a runtime shield, analogous to safe-RL shielding [47,48]. The literature on LLM verification and validation similarly emphasizes that model outputs should be subjected to explicit verification and runtime monitoring rather than treated as intrinsically trustworthy [49]. The contribution claimed here is the integration and measured evaluation of these elements in a data-center CPS, not a new estimator or shielding theory.

Table 1 makes the evidence boundary explicit. AICEBERG is the closest recent agentic comparator because it separates LLM-generated instrument requests from deterministic bounds, allow/deny, and health checks [45]. Its physical RF testbed is stronger than our hardware evidence, while it does not address deceptive-sensor consensus. RADICS provides a runtime-assurance switch to a safety controller on a physical CPS [46]; our process-local verifier is analogous in purpose but does not inherit its formal or isolation guarantees. Das et al. and Unity provide stronger physical and classic data-center MAS evidence, whereas our narrower contribution adds adversarial sensing and explicit typed command admission. These systems are therefore qualitative comparators, not interchangeable quantitative baselines.

Table 1. Qualitative comparison with adjacent multi-agent and runtime-assurance systems.

System	Coordination object	AI/LLM role	Execution boundary	Evidence and boundary
This work	thermal sensor reports; FPR–OWA and persistent CRP	optional reactive or language supervisor	process-local snapshot/action binding, deterministic gate, fallback	public-data replay and illustrative plant; no hardware, PKI/RBAC, or energy claim
AICEBERG [45]	RF instrument requests via MCP/SCPI	language intent to structured tool request	separate execution and safety/resource servers; allow/deny and bounds checks	physical RF instrument; different domain and evaluation unit
Das et al. [10]	data-center power and workload allocation	none	coordinated utility policies	physical BladeCenter; no deceptive-sensor consensus or LLM gate
Unity [11]	decentralized resource allocation	none	resource arbiter among autonomic elements	Linux-cluster prototype; safety shielding is outside its focus
RADICS [46]	intelligent CPS commands	advanced AI/ML controller	runtime-assurance switch to a safety controller	physical water-treatment testbed; its guarantees are not inherited here

2.3. Soft Consensus in Fuzzy Group Decision Making

The consensus layer in this paper draws on four decades of research on consensus reaching in fuzzy GDM. Its foundations include the calculus of linguistically quantified propositions [25], the OWA operator as the canonical implementation of relative quantifiers such as "most" [26], and the notion of a soft degree of consensus introduced by Kacprzyk and colleagues. In this formulation, consensus means that most of the important individuals agree on most of the relevant options [27–29]. Herrera et al. extended this view to linguistic assessments [30], while Bordogna et al. modeled linguistic consensus through OWA aggregation [50]. Fuzzy preference relations subsequently became the standard representation of graded pairwise judgments, supported by integration results across preference formats [51] and consensus models operating across heterogeneous structures [31].

Three subsequent developments are particularly relevant. First, consensus models incorporated algorithmic guidance: consensus degrees and proximity measures, computed at multiple representation levels, drive automatic feedback that identifies which participants should revise specific assessments [31,33]. Second, consistency control was integrated with consensus so that agreement would not be reached at the cost of self-contradictory preferences, including under incomplete information [32,35]. Third, the field produced systematic analyses and reviews of soft consensus models in fuzzy environments [34,36,52], consensus measures [53], minimum-adjustment feedback mechanisms [54], and dynamic consensus and opinion dynamics in networked settings [55]. Operator-level tools, including type-1 OWA for multi-granular linguistic contexts [56], and methodologies for unbalanced linguistic term sets [57], complement this line of work. Applications include collaborative knowledge engineering under linguistic fuzzy consensus [58].

This tradition treats consensus as a monitored process involving unequally reliable participants, mirroring the structure of a zone of sensor agents under deception. Its transfer to machine-to-machine settings, however, remains limited. Consensus-reaching models assume human experts who revise their opinions, whereas sensor agents require automated analogs of reliability, preference, dominance, and feedback. Constructing these analogs forms the methodological core of this paper.

2.4. LLM-Based Agents and Trustworthy Agentic Orchestration

LLM-based agents combine language-model reasoning with tool use and environment interaction [39]. ReAct interleaves explicit reasoning traces with actions, improving groundedness and auditability [37]; Reflexion adds verbal reinforcement, allowing agents to revise behavior in response to feedback without weight updates [38]. Reasoning-quality mechanisms such as chain-of-thought prompting [59] and self-consistency over sampled reasoning paths [60] establish a sampling-and-agreement intuition generalized by softer fuzzy notions of consensus. In safety-critical domains, however, reasoning traces alone do not establish trustworthiness; explicit verification stages and human-auditable evidence are required. We adopt this position architecturally: the LLM planner acts as a proposer, never as an executor, and a deterministic verifier governs the actuation decision.

2.5. Research Gap and Positioning

The literature provides hierarchical sensing [8,9], multi-agent data-center management [10,11], lightweight approximate agreement [15], a mature theory of monitored, feedback-driven consensus [36], and agentic orchestration patterns [37,38]. We are not aware of prior work that integrates these elements within a blockchain-independent data-center CPS architecture: the operational state is the object of consensus rather than a ledger; reporter trust is represented through fuzzy preference relations and quantifier-guided dominance; consensus is monitored and repaired over time; and LLM-driven control remains subject to deterministic verification. The contrast with blockchain-IoT consensus concerns the object of agreement rather than the quality of the mechanism. Ledger consensus establishes agreement on the order and integrity of transactions under Byzantine quorum bounds, whereas the problem addressed here concerns graded-trust estimation of the current physical state at telemetry rates. The two approaches are therefore complementary.

A ledger can notarize the validated state produced by our layer, but it is not an estimation baseline for the problem studied here.

3. Architecture and System Model

This section specifies the proposed architecture, the agents involved, and the system and fault assumptions adopted in the study.

3.1. Design Principles

The design follows five principles.

- **P1, state consensus.** The agreement mechanism validates the operational state at each decision round. It maintains no ledger and does not require a global ordering of events.
- **P2, lightweight aggregation.** Agreement is computed locally over zone-sized groups of reports using closed-form operators, namely medians, fuzzy preference relations, and OWA weights. This keeps the consensus step compatible with edge-class hardware.
- **P3, monitored process.** In line with the soft-consensus tradition [36], agreement is tracked through consensus, proximity, and consistency, then repaired through targeted feedback instead of being assumed after one aggregation step.
- **P4, bounded autonomy.** LLM-based planning is used only to generate proposals. In the tested process, the decision to actuate remains with a deterministic verifier; production non-bypassability would additionally require process isolation, authenticated identities, and deployment controls.
- **P5, traceability.** Every consensus snapshot, agent proposal, verifier decision, and refinement cycle is recorded in structured, replayable execution traces.

3.2. Layered Architecture and Agent Roles

Figure 1 and Table 2 represent the architecture as two interacting paths. The *fast path* is deterministic and runs at every control tick: sensing agents feed the edge fuzzy-consensus operator and CRP monitor; the resulting snapshot and a bound proposal are then checked by an actuation verifier, which releases either an admitted control action or a conservative deterministic fallback. The *slow path* is an event-triggered supervisory loop. A pluggable supervisory agent reasons over consensus snapshots and proposes occasional supervisory actions, but those proposals still return to the *same* verifier before they can affect actuation. A trustworthiness-and-traceability plane spans both paths.

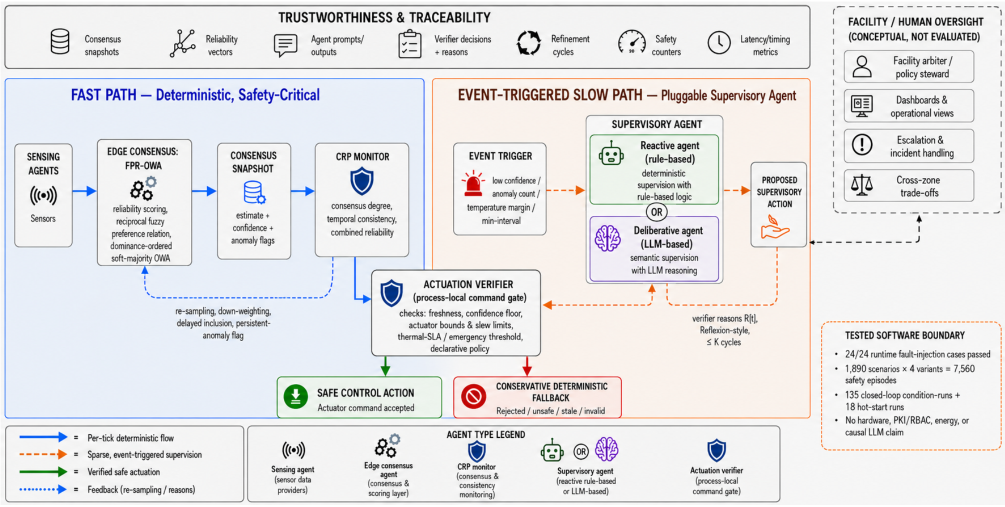

Figure 1. Numbered information flow and authority boundaries. The CRP monitor diagnoses agreement but cannot admit commands; the actuation verifier is the only process-local command-admission component. The optional supervisor is outside the deterministic per-tick path.

Table 2. Non-overlapping runtime roles and typed interfaces.

Component	Consumes	Produces	Authority boundary
Sensing agent	local measurement, timestamp, reporter ID	signed-in-process report record	Cannot aggregate or actuate
Edge consensus agent	report vector	FPR–OWA estimate, confidence, flags, reliability vector	Cannot admit commands
CRP monitor	current snapshot, persistent history	consensus degree, exclusions, feedback, updated snapshot	Diagnoses and requests re-sampling; does not decide actuation
Supervisory proposer	immutable snapshot and optional history	typed action proposal and rationale	Optional; has no actuator interface
Actuation verifier	bound snapshot–proposal pair, policy, actuator state	accept/reject reasons and one-shot authorization	Sole process-local command-admission point
Actuator/fallback	accepted authorization or rejection/timeout	bounded state change or conservative local action	Rejects missing/consumed authorization
Trace recorder	all identifiers, payloads, verdicts, timings	append-only experiment record	Observes; cannot change decisions

The end-to-end sequence is: (1) reporter records are collected; (2) the edge agent computes FPR–OWA and a typed snapshot; (3) the CRP monitor updates its persistent diagnostics and feedback; (4) the deterministic fast controller forms a local proposal, while an event may additionally request a supervisory proposal; (5) the actuation verifier validates snapshot schema, freshness, confidence, proposal binding, actuator bounds, slew, and the thermal policy; (6) a one-shot authorization changes actuator state, or rejection/timeout selects the local fallback; and (7) identifiers, payloads, reasons, state change, and timings are appended to the trace.

Physical and sensing layer

Sensor agents attached to racks, hosts, and power-distribution interfaces emit per-round observations of thermal, power, and workload variables, following hierarchical data-center monitoring practice [8,9]. Actuator interfaces (fan speed, cooling setpoints) close the physical loop.

Edge consensus layer

Edge moderating agents collect the reports for one zone and execute the FPR–OWA lightweight consensus operator (Section 4.3). The output is a zone state estimate enriched with diagnostic metadata: per-sensor reliability scores, anomaly flags, and an aggregate confidence value. This result is packaged as a *consensus snapshot*, which is the only sensing artifact exposed to the upper layers.

Zone consensus monitoring layer

CRP monitor agents follow the consensus process over time. They compute consensus degree, per-agent proximity, and temporal consistency, and trigger the adaptive feedback actions of Section 4.5 (re-sampling, down-weighting, delayed inclusion, persistent-anomaly flagging) when agreement degrades. For sensor agents, this layer plays the role of the moderator used in consensus-reaching models [32,33].

Slow-path agentic control

A pluggable supervisory agent reasons over consensus snapshots and proposes structured *supervisory* control actions. It may be a reactive (rule-based) agent or a LangGraph-orchestrated LLM agent under a ReAct-style scheme [37]; the comparison is reported in Section 7.4. In the intended event mode this layer is invoked sparsely when a configured policy fires on low confidence, anomaly flags, temperature margin, or a minimum inter-invocation interval. Historical records show that an LLM call may cost seconds, and frequent invocation reduces responsiveness without improving estimation. Every proposal is checked by the same verifier, which accepts it, rejects it, or returns machine-readable feedback for optional Reflexion-style self-refinement [38], limited to at most K cycles (Section 5).

Facility operation layer

A facility arbiter resolves cross-zone trade-offs, including thermal risk versus energy cost and workload placement, in line with the local–global decomposition of autonomic data-center management [10,11]. It also exposes dashboards for human oversight and escalation. In this study, the arbiter is kept as a conceptual, single-facility provision: the experiments operate at zone level and do not exercise it, while cross-zone arbitration is left to future work (Section 9).

Trustworthiness and traceability plane

Across all layers, a traceability service records consensus snapshots, reliability vectors, agent prompts and outputs, verifier decisions with reasons, refinement cycles, and safety counters. These records provide an auditable and replayable account of each control decision (P5).

3.3. System Model

Let $\mathcal{Z}$ be the set of zones. Zone $z \in \mathcal{Z}$ contains n_z sensing agents indexed by $E_z = \{1, \ldots, n_z\}$. Time is discretized into decision rounds $t = 1, 2, \ldots$ aligned with the telemetry period. At round t, agent $i \in E_z$ reports a measurement

$$s_i(t) \;=\; x_z(t) \;+\; \varepsilon_i(t) \;+\; a_i(t), \tag{1}$$

where $x_z(t)$ is the latent zone state (in the reference use case, a zone thermal variable), $\varepsilon_i(t)$ is zero-mean sensing noise, and $a_i(t)$ is an attack term, zero for honest reports. The exposition uses a scalar state for clarity; the operators extend componentwise to state vectors in $\mathbb{R}^d$ that combine thermal, power, and workload features.

Communication between sensing and edge agents is local, frequent, and loss-prone; messages may be delayed, dropped, or duplicated, in the spirit of lightweight agreement in lossy networks [15]. Zone-to-facility communication is less frequent and carries only consensus snapshots and verified control intents. No global clock or reliable transport is assumed. Each snapshot carries a timestamp, and the verifier checks it for staleness (Section 5.2).

3.4. Fault and Attack Model

We consider a deception adversary that compromises a fraction $0 \leq \beta_a < 1$ of the sensing agents of a zone and manipulates their reports through the attack term $a_i(t)$ in (1). Table 3 reports the attack taxonomy, chosen to cover the canonical false-data-injection patterns studied in CPS security [14] as they appear in slow-dynamics thermal telemetry. Benign faults, including missing and delayed reports or transient noise bursts, are handled by the same machinery. The consensus layer does not attempt to distinguish malice from malfunction: both appear as unreliable reports.

Table 3. Deception-attack taxonomy. A fraction β_a of a zone's agents is compromised, possibly intermittently.

Attack	Form of $a_i(t)$	Operational effect
Bias	constant offset b	Systematic over/under-reporting that shifts the zone estimate
Drift	ramp $\delta\,(t - t_0)$	Slow poisoning that mimics gradual thermal change and evades thresholding
Freeze	$s_i(t) \equiv s_i(t_0)$	Stale value masking real dynamics (replay of own last reading)
Replay	$s_i(t) = s_i(t - \Delta)$	Re-injection of a past window, plausible in distribution but desynchronised
Scaling	multiplicative $(\gamma - 1)\,x_z(t)$	Proportional distortion that preserves trends while corrupting magnitude
Mixed	time-varying combination	Alternating patterns that defeat single-statistic defences
Spike	repeated alternating impulses $\pm b$ over the active window	Abrupt transient corruption that stresses outlier rejection; this is a sequence of impulses, not a single isolated sample

The defender's goal is not exact agreement under arbitrary Byzantine behaviour. Ledger consensus can provide that guarantee, but at a cost incompatible with per-round telemetry. The target here is graceful degradation: a zone estimate whose error grows slowly with β_a and attack magnitude, while the implicated reporters are flagged for the feedback loop of Section 4.5.

4. Lightweight Consensus Methodology

This section specifies the consensus layer: the per-round FPR–OWA operator executed by edge agents (Section 4.3) and the zone-level consensus-reaching loop that monitors and repairs agreement over time (Section 4.5). In doing so, it adapts four ingredients of soft-consensus GDM to sensor agents: graded reliability replaces expert importance, fuzzy preference relations replace pairwise expert judgements, quantifier-guided dominance replaces selection degrees, and feedback replaces moderator advice.

4.1. Reliability Scoring

Let $\mathbf{s}(t) = (s_1(t), \ldots, s_{n_z}(t))$ denote the report vector of zone z at round t (zone subscripts are omitted where unambiguous). The edge agent first computes a robust center and scale:

$$m(t) = \mathrm{median}_i\ s_i(t), \qquad \hat{\sigma}(t) = 1.4826 \cdot \mathrm{MAD}_i\ s_i(t) + \epsilon, \tag{2}$$

where MAD is the median absolute deviation, the constant 1.4826 makes $\hat{\sigma}$ consistent with the standard deviation under Gaussian noise, and $\epsilon > 0$ prevents degeneracy when reports coincide. Each agent then receives a reliability score

$$\rho_i(t) \;=\; \exp\!\Big(-\,\frac{|s_i(t) - m(t)|}{\hat{\sigma}(t)}\Big), \qquad 0 < \rho_i(t) \le 1. \tag{3}$$

Reliability decays exponentially with robust-standardised distance from the group center. Agents close to the peer center receive scores near one, while outliers, whether noisy or attacked, are discounted smoothly rather than removed by a hard threshold. This is the sensor-agent analogue of the graded importance weights used in soft consensus [27,28].

4.2. Fuzzy Preference Relation over Reporters

Pointwise reliability only partly captures the relational structure of trust: ordering reporters requires assessing how much more credible one report is than another. Following the representation of graded pairwise judgements by fuzzy preference relations [31,51], we define the entries $p_{ij}(t)$ of $P(t)$, with $0 \le p_{ij}(t) \le 1$, by

$$p_{ij}(t) \;=\; \frac{1}{1 + \exp\big(-\kappa\,(\rho_i(t) - \rho_j(t))\big)}, \qquad p_{ii}(t) = 0.5, \tag{4}$$

where $\kappa > 0$ controls the sharpness of the comparison. The relation is additively reciprocal, $p_{ij} + p_{ji} = 1$, and $p_{ij} > 0.5$ means that report i is more credible than report j. The logistic link is insensitive to small reliability differences, which map to values near 0.5, while saturating in clear-cut cases.

From $P(t)$, each agent receives a dominance degree

$$\delta_i(t) \;=\; \frac{1}{n_z - 1} \sum_{j \ne i} p_{ij}(t) \;\in\; [0,1], \tag{5}$$

namely the average strength with which i is preferred to its peers. This is the operational counterpart of the quantifier-guided dominance degrees used to exploit fuzzy preference relations in GDM selection processes [31,51]; it induces the credibility ordering used by the aggregation step.

Proposition 1 (order preservation). Let $n_z \ge 2$, $\kappa > 0$, and let p_{ij} and δ_i be defined by Eqs. (4)–(5). For any reporters i and ℓ, $\rho_i > \rho_\ell$ if and only if $\delta_i > \delta_\ell$, and $\rho_i = \rho_\ell$ if and only if $\delta_i = \delta_\ell$.

Proof. Write $g(x) = (1 + e^{-\kappa x})^{-1}$. Since $\kappa > 0$, g is strictly increasing and $g(x) + g(-x) = 1$. If $\rho_i > \rho_\ell$, then, for every $j \notin \{i, \ell\}$, $g(\rho_i - \rho_j) > g(\rho_\ell - \rho_j)$; the remaining paired terms also satisfy $g(\rho_i - \rho_\ell) > g(\rho_\ell - \rho_i)$. Summing these $n_z - 1$ strict inequalities and dividing by $n_z - 1$ gives $\delta_i > \delta_\ell$. Interchanging i and ℓ proves the reverse direction. If $\rho_i = \rho_\ell$, corresponding summands are equal, including the paired term $g(0) = 1/2$, hence $\delta_i = \delta_\ell$; the strict result excludes equality when the reliabilities differ. □

The implementation extends exact computed-score ties by immutable reporter ID, yielding a deterministic linear order. This convention does not make the operator invariant to reporter renumbering and does not guarantee bitwise cross-platform equality for near-equal floating-point scores. The paper-facing audit found 50 legacy-estimate differences only where dominance gaps were at most 10^{-12}; affected displayed cells were regenerated.

4.3. Quantifier-Guided OWA Aggregation

Let σ be the permutation of E_z that sorts dominance degrees in non-increasing order, $\delta_{\sigma(1)} \geq \cdots \geq \delta_{\sigma(n_z)}$. Aggregation weights are derived from a regular non-decreasing linguistic quantifier in the parametric family of Zadeh [25],

$$Q_{a,b}(r) = \begin{cases} 0, & r < a, \\ \dfrac{r-a}{b-a}, & a \leq r \leq b, \\ 1, & r > b, \end{cases} \qquad 0 \leq a < b \leq 1, \tag{6}$$

instantiated with the weak-majority quantifier "most", for which we use $(a,b) = (0.3, 0.8)$ and $(0.2, 0.7)$ [30,31]; the released configuration labels this pair (α, β), but to avoid collision with the blend coefficient of Section 4.5 we write it (a,b) throughout. The base Yager weights [26] are $\omega_j = Q_{a,b}(j/n_z) - Q_{a,b}((j-1)/n_z)$, $j = 1, \ldots, n_z$, and are assigned to the dominance-ordered reports in *reverse*, so that the weight on the k-th most credible report is

$$\begin{aligned} w_k &= Q_{a,b}\Big(\tfrac{n_z-k+1}{n_z}\Big) - Q_{a,b}\Big(\tfrac{n_z-k}{n_z}\Big), \\ \hat{x}_z(t) &= \sum_{k=1}^{n_z} w_k \, s_{\sigma(k)}(t). \end{aligned} \tag{7}$$

Because $Q_{a,b}$ is constant outside $[a,b]$, the operator assigns *no* weight to the single most- and least-credible reports and concentrates the mass on the high-credibility majority between them. It therefore acts as a robust *soft-majority* over the dominance order, or equivalently as a graded, trust-induced analogue of a trimmed mean. Its core properties are stated explicitly to avoid ambiguity: (i) the preference relation is additively reciprocal, $p_{ij} + p_{ji} = 1$; (ii) the weights satisfy $w_k \geq 0$ and $\sum_k w_k = 1$, so $\hat{x}_z$ is a convex combination of the reports and lies in their range; (iii) the orness of w for "most" is ≈ 0.5, i.e. the fusion is near-median rather than optimistic or pessimistic; (iv) since the extreme ranks receive zero weight, no single reporter, however credible or however deviant, can set the estimate by itself, which bounds the influence of one compromised sensor; and (v) the operator costs $O(n_z^2)$ per zone and round (Section 4.4). Ordering is induced by dominance rather than by the report values, so the operator is an induced OWA whose order-inducing variable is relational credibility. This distinguishes it from a direct OWA on sorted measurement values, retained as an ablation baseline (Section 7.2).

4.4. Diagnostics and Consensus Snapshot

The edge agent emits diagnostics alongside the estimate. Agents whose reliability falls below an adaptive threshold are flagged as anomalous:

$$\begin{aligned} A_z(t) &= \big\{\, i \in E_z : \, \rho_i(t) < \tau_\rho(t) \,\big\}, \\ \tau_\rho(t) &= \max\!\big(\underline{\tau},\ \mathrm{median}_i\, \rho_i(t) - 2\ \mathrm{std}_i\, \rho_i(t)\big), \end{aligned} \tag{8}$$

with a floor $\underline{\tau}$ that keeps the threshold meaningful when reliabilities are uniformly high. The zone confidence is the mean reliability $\gamma_z(t) = \frac{1}{n_z} \sum_i \rho_i(t)$. The *consensus snapshot*

$$\chi_z(t) = \big(\hat{x}_z(t),\, \gamma_z(t),\, A_z(t),\, \boldsymbol{\rho}(t),\, P(t),\, t\big) \tag{9}$$

is the typed artefact passed upward. It carries enough evidence for the verifier to police staleness and confidence, and for the traceability plane to support post-hoc audit. The operator costs $O(n_z^2)$ arithmetic per zone and round, dominated by the preference relation. For the tested zone sizes this computation is sub-millisecond in the reported CPU environment and involves no inter-zone communication, unlike the $O(n^2)$ *message* complexity of PBFT-style quorums [24].

4.5. Zone-Level Consensus Reaching and Adaptive Feedback

One aggregation round, even a robust one, cannot distinguish a briefly noisy agent from a persistently deceptive one. Nor can it recover useful information from agents that are repeatedly down-weighted. The zone verifier therefore runs a consensus-reaching process (CRP) over rounds, adapting the guidance-and-feedback scheme of soft-consensus GDM [31–33,35] to sensor agents.

Consensus degree and proximity

Pairwise similarity between reports is measured by soft coincidence,

$$s_{ij}(t) = \max\left(0,\ 1 - \frac{|s_i(t) - s_j(t)|}{M_z}\right), \tag{10}$$

with $M_z > 0$ a zone normalisation constant (e.g., the admissible operating range of the monitored variable). Agent proximity and the zone consensus degree follow the three-level construction of consensus models [33]:

$$p_i(t) = \frac{1}{n_z - 1} \sum_{j \neq i} s_{ij}(t), \qquad C_z(t) = \frac{1}{n_z} \sum_{i \in E_z} p_i(t). \tag{11}$$

Temporal consistency

Mirroring the consistency control that prevents agreement from masking self-contradiction [32,35], each agent is also scored for coherence with its own recent behaviour. Let $\tilde{s}_i(t)$ be a short-horizon prediction of agent i's report, an exponentially weighted moving average of its past reports with smoothing $\lambda = 0.3$. The temporal consistency score is

$$k_i(t) = \max\left(0,\ 1 - \frac{|s_i(t) - \tilde{s}_i(t)|}{M_z}\right), \tag{12}$$

and the combined per-agent reliability used by the CRP blends social agreement and self-coherence with blend coefficient $\alpha \in [0, 1]$ (distinct from the quantifier bounds (a, b)),

$$r_i(t) = \alpha\, p_i(t) + (1 - \alpha)\, k_i(t). \tag{13}$$

Replay and freeze attacks motivate this combination. Because self-EWMA cannot represent a common zone trend, we also evaluate an aligned innovation score

$$\begin{aligned} k_i^{\mathrm{dyn}}(t) &= \max\left(0, 1 - \frac{|s_i(t) - \hat{x}_z^{\mathrm{dyn}}(t)|}{M_z}\right), \\ \hat{x}_z^{\mathrm{dyn}}(t) &= f(\hat{x}_z(t-1), T_a(t-1), L(t-1), u(t-1)), \end{aligned} \tag{14}$$

where the frozen first-order predictor uses the previous estimate, ambient temperature, load, and the action actually executed at the previous tick. True temperature is never an input. This alternative captures prediction innovation relative to common zone dynamics; Section 7.2 reports it as an offline diagnostic ablation rather than retroactively replacing the published CRP.

Acceptance and feedback

Given thresholds τ_c (zone consensus) and τ_r (agent reliability), the verifier proceeds as follows at each round:

1. compute $\chi_z(t)$ via Sections 4.1–4.4, and $C_z(t)$, $r_i(t)$ via (11)–(13);
2. if $C_z(t) \geq \tau_c$, accept and forward $\chi_z(t)$;
3. otherwise, build the feedback set $F_z(t) = \{ i : r_i(t) < \tau_r \}$ and apply to each $i \in F_z(t)$ one or more corrective actions: *re-sampling* (request a fresh report), *temporary down-weighting* (attenuate i's influence in the next aggregation), *delayed inclusion* (exclude i for one round), or a *persistent-anomaly flag* when r_i remains below τ_r for L consecutive rounds, which escalates the agent to the facility layer for inspection;
4. re-validate and forward the resulting snapshot, recording all feedback actions in the trace.

This is the sensor-agent analogue of moderator advice generation: the process targets the participants most responsible for disagreement instead of recomputing blindly over the whole zone [33,35]. The CRP has the same $O(n_z^2)$ per-round cost as the operator itself and never blocks the control loop. When consensus cannot be restored within a round, the snapshot is forwarded with degraded confidence, and the confidence gate of the verifier (Section 5.2) decides whether control may proceed. We evaluate the CRP at $\alpha = 0.5$, $M_z = 4.0$, $\tau_c = 0.7$, $\tau_r = 0.5$, $L = 3$ (Section 7.2); in the offline trace replay used there, the *re-sampling* action is realised as delayed inclusion/down-weighting of the implicated reporters, the only substitution the recorded data permits.

5. Verifier-Gated Agentic Control

The control side of the architecture maps validated state to safe actuation. It separates a flexible *proposer*, the supervisory agent, from a rigid deterministic *gatekeeper*. The supervisory role is *pluggable*: it may be filled by a reactive (rule-based) agent or by a deliberative LLM agent, while both remain subordinate to the same verifier. We first describe the LLM instantiation, then the reactive counterpart, and compare them empirically in Section 7.4.

5.1. Event-Triggered Supervisory Agent

The deliberative agent is an LLM orchestrated with LangGraph and backed, in our implementation, by a Gemma instruction-tuned model served behind a standard inference endpoint. Under a ReAct-style scheme [37], it receives the consensus snapshot $\chi_z(t)$ and, where relevant, recent history and the current actuator state. It then reasons explicitly about the zone's thermal and energy condition and emits a *structured* supervisory action

$$u_z(t) = \big(\Delta f,\ \theta_{\text{set}},\ \textit{rationale},\ \textit{used\_consensus}\big), \tag{15}$$

where Δf is a fan-speed adjustment, θ_{set} a cooling setpoint, *rationale* a natural-language justification retained for audit, and *used_consensus* a declaration that the proposal was grounded in the validated snapshot rather than in raw or remembered values. Output is constrained to a typed JSON schema. Malformed or schema-violating generations do not stall the loop; they trigger the deterministic fallback described below and are recorded as handled events, not crashes.

The agent is designed to remain outside the per-tick critical path. It can be invoked by an event-triggered policy parameterised by a confidence threshold, an anomaly-count threshold, a temperature margin, and a minimum inter-invocation interval. Between triggers, the fast path runs alone. The agent is also interchangeable: any instruction-following LLM can fill the role, and the architecture's *safety* properties depend on the verifier rather than on well-behaved agent outputs.

Low confidence has two intentionally different meanings: it requests diagnosis or supervisory escalation, but the admission rule rejects observation-dependent actuation at or below the confidence floor. Thus the trigger and gate do not conflict. Escalation can explain, alarm, or request new sensing, while the local fallback preserves liveness without trusting the low-confidence state. In the implementation both thresholds are 0.35. The new closed-loop pilot invokes deterministic supervision periodically at every tick and is not evidence for an event-trigger latency benefit.

Reactive supervisory agent

As a deterministic counterpart, we implement a reactive agent fired by the *same* triggers but emitting actions from fixed rules: a proportional fan/setpoint response to the thermal margin and anomaly state, with no language model. It passes through the identical verifier and fallback. This agent is the control baseline for RQ4, since it isolates what, if anything, LLM reasoning adds beyond a well-engineered rule under the same conditions.

Action-only invocation

The architecture does not rely on continuous LLM reasoning for safety-critical control. Per-tick control is handled by the deterministic fast path, while the LangGraph/Gemma supervisor is invoked only under event-triggered conditions and can only *propose* actions, never execute them. An action-only (or tool-only) invocation, where the supervisor is constrained to emit only the structured machine-readable action of Eq. (15) or a tool call without a free-text rationale, is therefore compatible with this design and would reduce token generation and latency. We treat this as an implementation-level optimisation: the tested behavior derives from deterministic consensus, verifier gating, and fallback control, and every supervisory output, whether verbose or action-only, must pass the same deterministic verifier. We do not benchmark it separately (Section 9).

5.2. Deterministic Verifier Agent

The verifier V evaluates every proposed action against a declarative policy through deterministic rules and contains no learned components. Its checks fall into two groups.

Process checks establish that the proposal is procedurally trustworthy: the agent declared grounding in the consensus snapshot; the snapshot is fresh, $t_{\text{now}} - t \leq \tau_{\text{age}}$; and the zone confidence clears the floor, $\gamma_z(t) \geq \tau_\gamma$.

Safety checks establish that the action is physically admissible: actuator values remain within configured bounds, slew rates stay within limits, and the predicted post-action zone temperature, obtained from a conservative first-order thermal model, complies with the service-level threshold T_{SLA}, with a stricter emergency rule activating at T_{emerg}.

The verdict is

$$V\big(u_z(t), \chi_z(t)\big) \in \{\text{ACCEPT}, \text{REJECT}\}, \text{ reasons } R(t), \tag{16}$$

where $R(t)$ is a machine-readable list of violated checks. Only ACCEPTed actions receive a monotonic one-shot authorization that the process-local actuator interface consumes once. The runtime copies and hashes the authoritative snapshot before callbacks, binds proposal and authorization to its zone, timestamp, request ID, and payload, and rejects stale, mutated, superseded, missing, or consumed authorizations. These checks validate command integrity inside the tested process. They are not user or agent authentication, RBAC/IAM, PKI, sandboxing, or proof of cross-process non-bypassability.

Architectural gating (not a formal proof)

The actuation path is wired exclusively through V: agent outputs have no side effects, and the actuator interface consumes only verifier-cleared actions. Liveness is preserved by a deterministic fallback policy, a conservative rule-based controller that takes over when no proposal is accepted within the refinement budget. Safety therefore does not depend on the agent behaving well, and control does not stall when the agent behaves badly. We use the terminology carefully: this is an *architectural* enforcement argument supported by empirical evidence (Section 7.3, including adversarial and failure-mode tests), not a formal proof of non-bypassability or of policy completeness. We therefore claim that the verifier *blocked all unsafe and process-invalid actions observed in the tested scenarios*, and treat a machine-checked guarantee as future work (Section 9).

5.3. Self-Refinement Cycles

On REJECT, the reasons $R(t)$ are returned to the planner, which may revise its proposal through a Reflexion-style verbal-feedback loop [38], for at most K cycles:

$$u_z^{(c+1)}(t) = \pi_{\mathrm{LLM}}\big(\chi_z(t),\, R^{(c)}(t)\big), \qquad c = 0, \ldots, K-1. \tag{17}$$

Refinement turns the verifier from a pure filter into a teacher: within the episode, the planner receives the violated constraint and can adjust its proposal without any weight update. The evaluation (Section 6) treats the planner variants, deterministic policy only, ReAct without verifier, ReAct with verifier, and ReAct with verifier and self-refinement, as ablation conditions for RQ3.

5.4. Trustworthiness and Traceability

Every round appends the following elements to the execution trace: the consensus snapshot and its diagnostics, CRP feedback actions, agent prompt and raw output, verifier verdict and reasons, refinement cycles, the executed action, and measured timings. From these traces, the framework derives the safety indicators used in the evaluation: rates of unsafe actions *proposed*, *blocked*, and, most critically, *executed*, together with acceptance rates and refinement-cycle counts. The same traces also support latency and overhead profiling. The trace design follows the auditability rationale of structured reasoning traces [59] and agreement-based stability signals over sampled outputs [60], but anchors trust in recorded *decisions* rather than in the persuasiveness of generated explanations.

Table 4. End-to-end pseudocode of the implemented per-tick pipeline.

Step	Operation
1	Collect typed reporter records S_t; reject invalid schema, zone, time, or non-finite fields.
2	Compute robust centre/scale, ρ_i, P, and δ_i; sort by $(-\delta_i, \mathrm{stableID}_i)$ and emit FPR–OWA snapshot χ_t.
3	Update persistent CRP state; compute C_t, proximity, temporal score, exclusions, and feedback; forward a converged or explicitly degraded snapshot.
4	Compute the local deterministic proposal. If the event policy fires, issue one immutable supervisory request; a late response retains its original request ID and may be superseded.
5	Bind the selected proposal to χ_t; check schema, freshness, confidence, declared grounding, actuator bounds/slew, and conservative thermal policy.
6	If accepted, emit and consume a one-shot authorization and update the actuator; otherwise execute the bounded local fallback or alarm.
7	Record identifiers, payloads, reasons, source, before/after state, CRP actions, and timings.

6. Experimental Setup

We evaluate the architecture through *simulation over publicly available telemetry derived from real systems*: a hierarchical IoT/MAS data-center testbed in which sensing agents, edge moderating agents, and the verifier-gated orchestrator operate over audited public datasets.

This wording is used throughout instead of "real-data evaluation". The study is driven by real datasets, but it is not a physical deployment, and the implications of that choice are discussed in Section 9. Table 6 collects all reported parameter values. As a reproduction gate, every headline number in the manuscript was recomputed from the released artifacts (Table 7).

6.1. Data and Their Audited Roles

Dataset names were treated as hypotheses rather than accepted at face value. Before use, each dataset was checked by an automated audit and admitted only for the role supported by its schema (Table 5). Thermal calibration, namely the hot-corridor temperature signal estimated by the consensus layer, comes from the Kaggle *Data center Warm/Hot Corridor Temperature Prediction* dataset (fields `TLHC`, `T_MEAS-*`, `P_cu-*`). The ENACT edge–cloud continuum telemetry (Zenodo record 18920397; `CPU%`, `MEM`, `Energy W`) is used as a power/workload proxy in the simulation (240 rows per scenario, recorded in each trace's metadata). The audit marks ENACT as usable for power but *not* for temperature, so it is never used as a thermal signal. Because the reported error metrics evaluate the thermal estimate, ENACT enters the experiments as simulation context rather than as an evaluated target. Anomaly and deception *timing* is taken from CuCD-ID, a CubeSat cybersecurity intrusion-detection dataset (Mendeley Data, CC BY 4.0). Under the audit's mapping rule, this cyber dataset contributes only the temporal placement of anomaly/attack episodes and is never interpreted as a thermal measurement. Google Cluster Data, Building Data Genome Project 2, and CICIoT2023 were considered as candidates or substitutes but were not used in the final run. The Kaggle thermal values appear standardised; all error metrics are therefore reported in dataset units unless inverse-scaling metadata is available.

Table 5. Datasets and their audited roles; each is used only for the role its schema supports.

Dataset	Source	Role	Temp.	Work./Pow.	Attack timing	License
Data Centre Warm/Hot Corridor Temperature Prediction	Kaggle	Thermal calibration (`TLHC`, `T_MEAS-*`, `P_cu-*`)	✓	—	—	Kaggle dataset terms
ENACT Edge–Cloud Continuum Telemetry	Zenodo rec. 18920397	Workload / power context (`CPU%`, `MEM`, `Energy W`)	—	✓	—	Record-specific
CuCD-ID (CubeSat Cybersecurity, Intrusion Detection)	Mendeley Data	Anomaly / deception *timing* only	—	—	✓	CC BY 4.0

6.2. Scenario Design

The replay campaign comprises 1,890 indexed scenarios. The historical safety archive contains four separately executed control variants for every scenario, hence $1{,}890 \times 4 = 7{,}560$ primary safety episodes. These are not four selected decision points within a trajectory: each row is one scenario–variant episode, and no event-trigger subset was used to obtain the 7,560 denominator. Triggered ticks belong to the separate historical realtime logs. Scenarios vary seven deception/anomaly attack types (Table 3: bias, drift, freeze, mixed, replay, scaling, spike), the attacked-sensor fraction $\beta_a \in \{0.0, 0.1, 0.2, 0.3, 0.4\}$, sensor noise levels $\{0.1, 0.25, 0.5\}$, zone sizes $n_z \in \{5, 9, 13\}$, quantifier bounds $(a, b) \in \{(0.2, 0.7), (0.3, 0.8)\}$, and three random seeds $\{11, 42, 101\}$. The thermal response follows a first-order model with disturbance, and attacks are active only over a bounded fraction of each episode rather than throughout the whole run. The seven attack types match Table 3; spike is implemented as repeated alternating impulses over its active window.

6.3. Consensus Operators and Supervisory Agents

Six aggregators are compared under identical conditions: arithmetic *mean*, *trimmed mean*, *Hampel-filter-then-mean*, *median*, *direct OWA* (value-ordered, same quantifier), and the proposed *FPR–OWA* (dominance-ordered, with reliability diagnostics). The monitored *CRP* of Section 4.5 is evaluated on top of FPR–OWA. For diagnostic validation, we also score three simple detectors, median+MAD z-score, Hampel residual, and trimmed residual, as baselines for the anomaly signal (Section 7.2). On the control side, four variants instantiate the safety ablation: *deterministic policy only*, *ReAct without verifier*, *ReAct with verifier*, and *ReAct with verifier and self-refinement*. A *reactive (rule-based) supervisory agent* is also present in the historical archive. Those reactive and LLM records were produced by non-identical execution paths, so Section 7.4 treats them as descriptive rather than a paired control experiment. Two historical supervision regimes are reported (Section 7.5): an *intensive* regime, which invokes the agent aggressively, and an *event-triggered* regime (confidence 0.35, anomaly 3, temperature margin 0.02, minimum interval 20 ticks, no real-time refinement). Both are evaluated against deadline budgets of $\{100, 250, 500, 1000, 5000\}$ ms.

6.4. Metrics

Estimation quality is measured by mean absolute error (MAE), root mean square error (RMSE), median absolute error, the 95th-percentile absolute error (P95 AE), and maximum absolute error, all in dataset units. We also report the mean number of anomaly flags and the mean confidence emitted by each method. Safety is measured by the counts of unsafe actions *proposed*, *blocked* by the verifier, and *executed*, the last being the critical count, together with accepted safe actions and LLM/verifier call counts. The anomaly signal is validated against ground-truth attacked-sensor labels using AUROC, AUPRC, precision/recall/F1, false-positive rate, and detection delay. Confidence is assessed through a reliability diagram, expected calibration error, and confidence–error correlation. Across-method comparisons use bootstrap confidence intervals, a Friedman test with Wilcoxon post-hoc, and a variance decomposition (η^2). Latency is reported per decision for the operators and per agent+verifier invocation for the control variants. The real-time study also reports loop-latency percentiles, invocation and failure rates, deadline-miss rates, and thermal-SLA violation rates. The additional action-dependent pilot reports integral absolute error (IAE) to the declared reference, time and integral above the SLA, peak excess, sustained target-band return onset, normalized fan effort, fan total variation, verifier rejections, fallback ticks, and CRP non-convergence. Normalized fan effort is not electrical energy. Replay and control pilots use three seeds $\{11, 42, 101\}$ and are descriptive; the estimation robustness extension uses ten seeds.

6.5. Implementation

The fast path is implemented in Python with vectorized NumPy operators. The orchestrator is built on LangGraph; the deliberative agent is a Gemma 12B instruction-tuned model served behind an OpenAI-compatible endpoint, while the reactive agent is a dependency-free rule engine. The verifier is also a dependency-free rule engine driven by a declarative SLA policy (T_{SLA}, T_{emerg}, τ_{age}, τ_{γ}, actuator bounds and slew limits, refinement budget K). All runs are seeded, and every scenario emits a full execution trace. These traces support exact replay and the post-hoc derivation of all reported metrics from measured files only.

Table 6. Reported parameter values, grouped by component (fixed across the campaign unless swept).

Component / symbol	Value
FPR sharpness κ	10.0
Anomaly floor $\underline{\tau}$	0.15
OWA quantifier (a, b)	$\{(0.2, 0.7), (0.3, 0.8)\}$
CRP blend α	0.5
CRP normalisation M_z	4.0
CRP thresholds τ_c, τ_r; persistence L	0.7, 0.5, 3
EWMA smoothing λ	0.3
Thermal model ρ, η, lag, σ_d	0.82, 0.018, 2, 0.08
SLA / emergency temperature	32 °, 38 ° (dataset units)
Verifier confidence floor τ_γ	0.35
Max consensus age τ_{age}	180 s
Fan / setpoint bounds (slew)	$[0, 1]$ (0.2), $[18, 28]$ (2.0)
Event trigger (conf., anomaly, margin, interval)	0.35, 3, 0.02, 20 ticks
LLM backend	Gemma 12B-it (OpenAI-compatible)

7. Results

We report only measured results, recomputed from the released artifacts. All 21 headline quantities pass the reproduction gate (Table 7). Throughout the section, "best" refers to the minimum-error method within the stated slice; no aggregate is projected or smoothed.

Table 7. Reproduction gate: headline quantities recomputed from the archived artifacts (representative subset; 21/21 pass).

Quantity	Reported	Recomputed	Status
Global MAE, mean	1.5366	1.53661	✓
Global MAE, FPR–OWA	1.6171	1.61713	✓
P95 AE, direct OWA	2.3906	2.39060	✓
Bias MAE, FPR–OWA	1.6468	1.64676	✓
Redundancy MAE, n_z=13	1.1522	1.15218	✓
Unsafe executed, no verifier	4409	4409	✓
Unsafe executed, verifier	0	0	✓
Event-triggered invocation rate	0.0396	0.03958	✓
Event-triggered deadline miss	0.0396	0.03958	✓
Event-triggered SLA violations	0.0	0.0	✓

7.1. RQ1: How the Consensus Operators Compare

Across the 1,890 scenarios, the six aggregators are close and *no method dominates* (Table 8). The plain *mean* achieves the lowest global MAE (1.5366) and RMSE (1.6210), with trimmed mean and Hampel-mean within 0.01; FPR–OWA is the highest-error method in this aggregate view (1.6171). A Friedman test over the six aggregators rejects strict equality ($\chi^2 = 1612.8$, $p < 10^{-3}$), and Wilcoxon post-hoc tests confirm mean<FPR–OWA on MAE ($p < 10^{-90}$). The practical effect is small, however: bootstrap 95% intervals overlap heavily, and the operator accounts for only $\eta^2 \approx 0.01$ of MAE variance. The ranking changes in the tail. On P95 AE, *direct OWA* (2.3906) and *median* (2.3952) lead the mean (2.4419), indicating that quantifier-guided aggregation helps in the worst case, where it is intended to help, even when it trails on average.

Table 8. Global estimation over all 1,890 scenarios (dataset units; lower is better; best per column in bold). Only FPR–OWA emits anomaly flags and a non-degenerate confidence (validated in Table 11).

Method	MAE	RMSE	Median AE	P95 AE	Max AE	Anomaly flags	Confidence
Mean	**1.5366**	**1.6210**	**1.4226**	2.4419	2.8048	–	1.000
Trimmed mean	1.5382	1.6216	1.4255	2.4370	2.7936	–	1.000
Hampel+mean	1.5464	1.6247	1.4330	2.4044	2.8681	–	1.000
Direct OWA	1.5668	1.6431	1.4619	**2.3906**	2.7126	–	1.000
Median	1.5925	1.6703	1.5063	2.3952	**2.6942**	–	1.000
FPR–OWA	1.6173	1.7073	1.5438	2.4849	2.8347	1.109	0.532

The advantage of a method also depends on the attack regime (Fig. 2). FPR–OWA is the best single aggregator under *bias* (1.6468 vs. mean 1.7763); Hampel-mean leads under *spike* and *mixed*; mean leads under drift, freeze, replay, and scaling; and at the heaviest compromise ($\beta_a = 0.4$), the *median* becomes most competitive (1.6133). MAE degrades smoothly with the attacked fraction; Fig. 3 shows the regenerated stable-ID FPR–OWA series. The main driver of accuracy is not the operator, but sensor redundancy (Table 9): a variance decomposition attributes $\eta^2 \approx 0.75$ of MAE variance to n_z, compared with 0.05 for attack type and 0.01 for the operator. Increasing the zone from 5 to 13 sensors reduces mean MAE from 1.8268 to 1.1522 (37%). RQ1 therefore resolves as follows: estimation quality is governed first by redundancy, then by attack regime, and only marginally by the aggregator. The fast path should match the operator to the threat rather than rely on a single fixed "winner". The (a, b) quantifier setting barely changes FPR–OWA (global MAE 1.610 vs. 1.624), whereas it moves the value-ordered direct OWA from 1.445 to 1.689; dominance ordering thus makes the fusion less sensitive to its own parametrization.

Table 9. MAE vs. sensors per zone (mean over attacks, ratios, noise).

Sensors/zone	Mean	Trimmed	Hampel	Direct OWA	Median	FPR–OWA
5	1.8268	1.8268	1.7851	1.8086	1.7752	1.8084
9	1.6309	1.6309	1.6987	1.7120	1.6907	1.7517
13	1.1522	1.1568	1.1554	1.1798	1.3116	1.2919

Robustness check. We repeated the estimation study over *ten* seeds (6,300 scenarios) with three additional robust baselines: a robust (Huber-gated) Kalman filter and Huber and Tukey M-estimators (Table 10). The conclusions do not change, and the uncertainty narrows. The per-seed standard deviation is ≈ 0.007–0.009, an order of magnitude below the between-method gaps, so the ranking is stable; mean still minimizes global MAE (1.538), while FPR–OWA still does not (1.614). Like the quantifier operators, the added robust estimators do not win on average but perform well in the tail: robust Kalman obtains the best P95 AE (2.346), and Huber the second best (2.365), compared with mean's 2.444. All operators remain lightweight ($\leq$ 0.086 s per scenario). The safety and control archives retain three seeds and are interpreted descriptively.

Table 10. Robustness check: estimation over ten seeds (6,300 scenarios) with added robust-Kalman and Huber/Tukey baselines (best per column in bold).

Method	MAE ± std	RMSE	P95 AE
Mean	**1.538** ± 0.008	**1.623**	2.444
Trimmed mean	1.539 ± 0.008	1.623	2.439
Hampel+mean	1.549 ± 0.007	1.628	2.407
Huber M-estimator	1.549 ± 0.007	1.623	2.365
Tukey M-estimator	1.554 ± 0.007	1.630	2.384
Direct OWA	1.567 ± 0.007	1.644	2.389
Median	1.591 ± 0.009	1.670	2.393
Robust Kalman	1.598 ± 0.009	1.668	**2.346**
FPR–OWA	1.614 ± 0.007	1.705	2.475

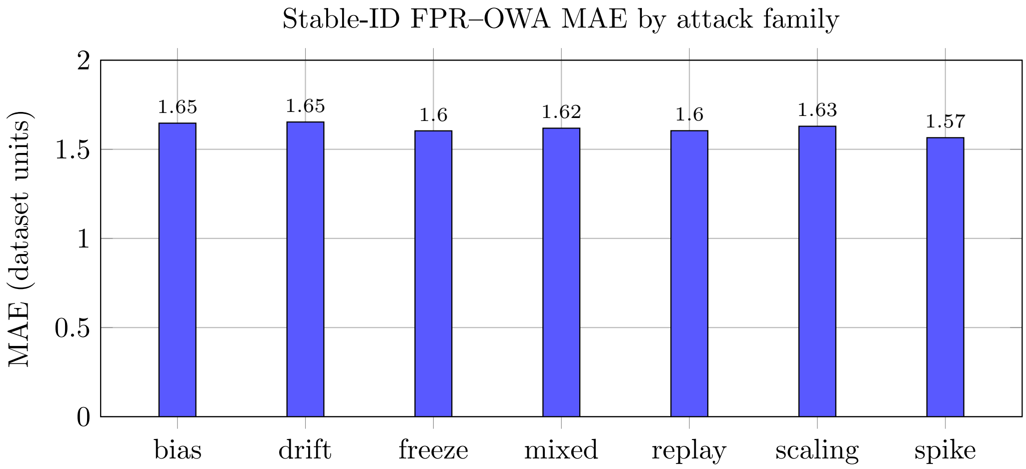

Figure 2. Stable-ID FPR–OWA MAE by attack family. Values were regenerated after the deterministic reporter-ID tie policy was applied.

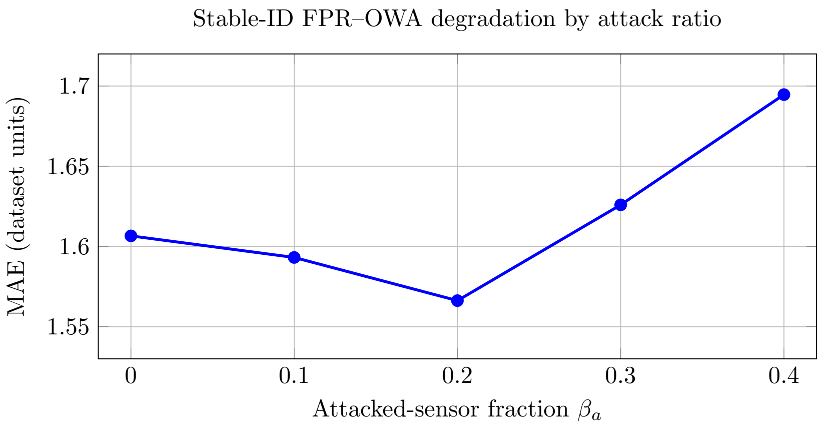

Figure 3. Stable-ID FPR–OWA MAE vs. attacked-sensor fraction β_a.

7.2. RQ2: Validating the Diagnostic Signal

The next question is whether FPR–OWA's trust signal justifies the accuracy penalty. We assess this directly against ground-truth attacked-sensor labels. Two results stand out (Table 11). First, the *single-round* FPR–OWA anomaly flag is *not* a stronger detector than a simple robust residual score. Its mean AUROC over attacked scenarios (0.649) is essentially the same as that of a median+MAD z-score (0.649) and slightly below the Hampel residual (0.658), while its false-positive rate is the highest in the group (0.104 vs. ≈ 0.04). What it does provide is an integrated confidence and anomaly signal, whereas plain aggregators emit a degenerate confidence of 1.000 and require an external detector.

The contribution appears when the monitored consensus-reaching process (CRP) of Section 4.5 is added. Temporal consistency and feedback make the diagnostic signal the most reliable one in the comparison: mean AUROC rises to 0.690, the false-positive rate falls by more than an order of magnitude to 0.005, and F1 is the best reported (0.273). Per-attack AUROC improves on six of seven families (Fig. 4), especially on bias (0.973 vs. 0.925), mixed (0.722 vs. 0.673), and the temporal attacks drift, replay, and scaling. The *freeze* attack remains difficult for every method (AUROC ≈ 0.4). The CRP also yields a small estimation gain, mostly in the tail (P95 AE 2.423 vs. 2.485; max 2.775 vs. 2.835), and converges quickly: 1.1 rounds on average, reaching the consensus threshold τ_c on 79.5% of ticks and excluding ≈ 0.2 reporters per tick (Table 12). On confidence calibration, the FPR–OWA score is correctly oriented but weak (Spearman $\rho = -0.23$ between confidence and absolute error; expected calibration error up to 0.12 in high-confidence bins), again comparable to residual-based confidences. RQ2 is therefore answered at the process level: diagnostic value comes from FPR–OWA+CRP, not from the single-round operator alone, and is most useful as a low-false-alarm trigger for verification and supervision, precisely the inputs consumed by the verifier and the event-trigger policy. An ablation of the operator's parts confirms the division of labour: *OWA-only* achieves the lowest MAE (1.580), while the reliability/dominance machinery supplies the trust structure rather than the accuracy gain (full operator MAE 1.616).

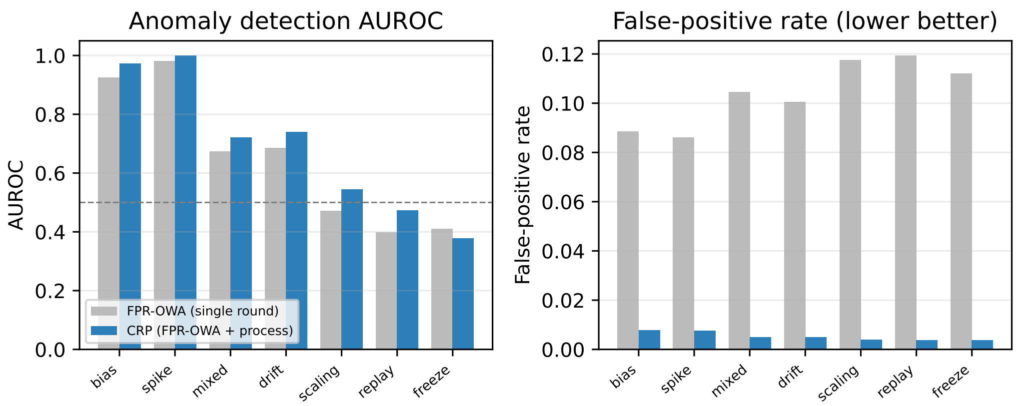

Figure 4. Anomaly detection by attack: AUROC (left) and false-positive rate (right), single-round FPR–OWA vs. the CRP.

Table 11. Anomaly-detection quality vs. ground-truth labels, averaged over attacks at $\beta_a > 0$ (sensor–tick level; best per column in bold).

Diagnostic source	AUROC	AUPRC	F1	FPR
Median + MAD z-score	0.649	**0.283**	0.234	0.044
Hampel residual	0.658	0.282	0.234	0.041
Trimmed residual	0.647	0.255	0.208	0.043
FPR–OWA (single round)	0.649	0.276	0.247	0.104
FPR–OWA + CRP	**0.690**	0.239	**0.273**	**0.005**

Table 12. CRP vs. single-round FPR–OWA on the same 1,890 scenarios (α_{crp}=0.5, M_z=4.0); convergence statistics at right.

Operator	MAE	P95 AE	Max AE	CRP statistic	Value
FPR–OWA (single round)	1.6173	2.4849	2.8347	Mean rounds to converge	1.10
FPR–OWA + CRP	**1.6116**	**2.4229**	**2.7753**	Converged within τ_c	79.5%
				Mean consensus degree C_z	0.772
				Mean reporters excluded/tick	0.20

Table 13. Temporal-consistency ablation on 45 frozen trajectories (10,800 ticks). Actions and plant trajectories are fixed; results are estimator diagnostics.

CRP reliability variant	MAE	Non-converged ticks	Exclusions	Rounds
Temporal only (sensor EWMA)	0.084507	637	1,142	11,181
Social proximity only	**0.079962**	57	2,826	11,763
Published blend	0.084417	627	1,171	**11,197**
Social + aligned zone innovation	0.079996	**22**	2,973	11,796

The temporal ablation in Table 13 directly tests the reviewed limitation of self-EWMA consistency. Adding the aligned zone-dynamic innovation reduces non-converged ticks from 627 to 22 relative to the published blend, but it increases exclusions and total rounds and has slightly worse MAE than social proximity alone (0.079996 vs. 0.079962). The outcome is therefore mixed. Because all four variants replay the same 45 trajectories and executed actions, it supports a diagnostic comparison only; it does not show that the dynamic variant improves counterfactual closed-loop control.

7.3. RQ3: Safety Episodes and Runtime Fault Injection

The historical safety archive contains four control variants for each of the 1,890 indexed scenarios, yielding 7,560 scenario–variant episodes (Table 14). These are complete offline safety episodes, not four selected trigger times per scenario. In the archived records, the unverified ReAct variant executed 4,409 unsafe proposals; the verifier-gated variants recorded zero unsafe executions and the refinement variant recorded more accepted proposals at the cost of additional calls. Because the historical variants do not all share the corrected runtime, these contrasts remain observational and do not identify the verifier as the sole cause.

The corrected runtime contract was tested directly through 24 controlled cases and 34 events. Invalid snapshots could not alter actuator state; proposals were bound to immutable snapshot/request identities; pending, timeout, supersession, mutation, and one-shot authorization cases behaved as specified. A separate nine-proposal adversarial set and failure modes for timeout, exception, malformed output, and missing authorization all selected rejection or bounded fallback. Across the tested records no unsafe or process-invalid action reached the simulated actuator. This is a bounded software observation plus an architectural enforcement argument. It is not a formal proof of policy completeness, physical safety, or non-bypassability in a deployed system.

Table 14. Safety outcomes (campaign totals), LLM supervisory agent. Without the verifier every unsafe proposal executes; with it, none do. ("No verifier" row: verifier calls are offline labelling calls.)

Agentic variant	Unsafe prop.	Unsafe blocked	Unsafe exec.	Safe acc.	LLM calls	Verifier calls
Deterministic policy only	0	0	0	7560	0	0
ReAct, no verifier	4409	0	4409	3151	7560	7560
ReAct + verifier	4358	4358	0	3202	7560	7560
ReAct + verifier + refinement	9127	9127	0	4341	13468	13468

7.4. RQ4: What Can the Historical Supervisor Records Establish?

Table 15 and Fig. 5 summarize the historical reactive and LLM records. They show equal stored MAE and no stored SLA or unsafe-execution event, with much lower recorded latency for the reactive path. A provenance audit, however, established that the two agents used non-identical execution paths. The records are therefore descriptive and cannot isolate supervisor type, verifier effect, or event triggering as a cause. The corrected common runtime is instead validated by 24 deterministic fault-injection cases and a 240-tick wiring smoke test: stale, future, wrong-zone, non-finite, malformed, mutated, superseded, missing, and consumed inputs were rejected or handled by the local fallback, while independently checked actuator bounds and slew held. This establishes software behavior for the tested contract, not a causal LLM comparison or production isolation.

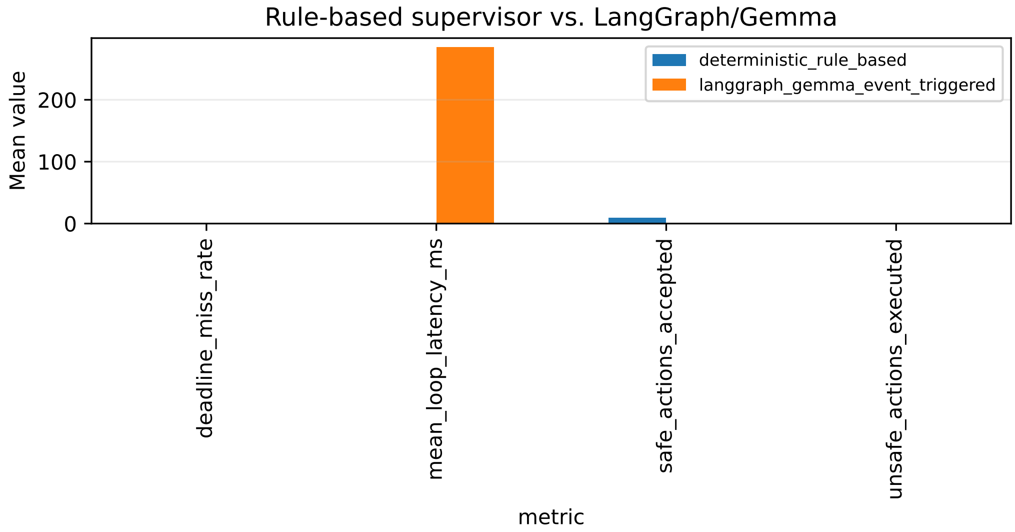

Figure 5. Descriptive historical reactive and LLM records. Because execution paths were non-identical, differences are not interpreted causally.

Table 15. Descriptive historical reactive and deliberative (LLM) supervisor records. Execution paths were non-identical; differences are not a paired causal comparison (means per archived scenario).

Metric	Reactive agent	LLM agent
Mean loop latency (ms)	**0.18**	285.4
P99 loop latency (ms)	**0.22**	7313.1
Deadline-miss rate	**0.0**	0.0396
Thermal-SLA violations	0.0	0.0
Unsafe actions executed	0	0
Accepted supervisory actions (mean/scenario)	9.5	0.0
Supervisor failure rate	0.0	0.011
MAE	1.5090	1.5090

7.5. RQ5: Computational Cost and Closed-Loop Behavior

The true persistent-CRP benchmark separates a convergent fresh-state input (one achieved round) from a frozen stress input (three achieved rounds) for $n_z = 5$–100. Median latency remains 140.1–271.4 μs for the convergent profile and 166.6–215.0 μs for the stress profile; median traced peak memory reaches about 244 KiB at $n_z = 100$ (Table 16). Each row uses 20 warm-ups, 500 latency repetitions, and 20 separate memory traces with tracing disabled during timing. The algorithmic work and preference storage are $O(n_z^2)$ per round, but six noisy timing points do not establish an empirical scaling law. These measurements contain neither electrical energy nor LLM-token consumption.

Table 16. True persistent-CRP benchmark (500 latency repetitions; 20 separate memory traces). C: convergent one-round input; S: frozen three-round stress input.

n_z	Median latency (μs)		P95 latency (μs)		Median peak memory (KiB)	
	C	S	C	S	C	S
5	140.1	166.6	153.6	185.4	10.3	10.5
9	141.1	180.2	152.4	194.0	11.2	11.2
13	147.6	180.1	169.3	199.1	12.3	11.4
25	148.2	187.8	165.1	209.2	19.5	18.4
50	184.2	198.2	208.5	219.8	65.4	63.7
100	271.4	215.0	297.8	238.7	244.1	241.3

Historical LLM decisions required seconds and motivate keeping language reasoning outside the critical path, but the intensive and event-triggered archives are retained only as non-paired descriptive evidence. In particular, the new action-dependent experiment used periodic deterministic supervision at every tick and cannot validate asynchronous live LLM behavior or a causal 94.7% event-trigger benefit.

The separate closed-loop pilot executes an illustrative first-order plant for 135 condition-runs (45 per controller), 240 ticks each. Table 17 adds standard control metrics. Mean IAE is 61.485 for the oracle, 69.326 for FPR–OWA+CRP, and 98.708 for mean; fan effort and total variation are reported separately. All original runs remained below the 32 °C SLA, so SLA exposure is zero and does not discriminate controllers. FPR–OWA+CRP also produced 627 non-converged ticks and 39 verifier rejections/fallbacks, and identified at least one attack-window alarm in only 14 of 42 attacked runs. These adverse outcomes are retained.

Table 17. Action-dependent illustrative thermal pilot: means over 45 condition-runs per controller (three seeds). Effort is normalized command-minutes, not electrical energy.

Controller	IAE	Estimator MAE	Fan effort	Fan variation	SLA exposure	Return onset (min)
Oracle	61.485	0.0000	135.247	2.420	0.000	1.156
Mean	98.708	0.2148	138.620	3.047	0.000	7.267
FPR–OWA+CRP	69.326	0.0844	135.984	3.164	0.000	6.178

The pre-specified hot-start appendix tests the missing unsafe-temperature boundary (Table 18). Starting from 33 °C, all clean systems recover sustainably below the SLA at minute 3. Under a common −3 °C measurement bias, the mean and FPR–OWA+CRP controllers recover at minute 68, eight minutes after bias removal, with about 50 °C-min of integrated excess; the oracle remains paired with its clean trajectory. This is evidence of dependence on common measurement quality, not robustness or energy saving.

Table 18. Exploratory hot-start boundary (three seeds, initial temperature 33 °C). Common bias is −3 °C at ticks 0–59.

Case	Controller	IAE	SLA exposure (min)	Integrated excess	Sustained recovery (min)
Clean	Oracle/Mean/FPR–CRP	—	3.000	1.919	3
Common bias	Oracle	111.664	3.000	1.919	3
Common bias	Mean	242.286	61.333	50.275	68
Common bias	FPR–OWA+CRP	242.678	61.333	50.073	68

8. Discussion

What the evidence supports

The main replay does not identify a universally best estimator: mean has the lowest global MAE, while sensor count explains most accuracy variation. The specific value of FPR–OWA+CRP is an integrated estimate and low-false-alarm diagnostic. The temporal ablation sharpens this interpretation. Zone-dynamic innovation improves convergence relative to the published blend, but social proximity alone has marginally lower MAE and the dynamic variant performs more exclusions and rounds. The appropriate conclusion is a trade-off, not general superiority.

The runtime results support separation of authority. The CRP monitor diagnoses agreement; the supervisor proposes; the actuation verifier admits a bound command; fallback preserves liveness. This resembles AICEBERG's separation of language reasoning from deterministic instrument checks and RADICS's runtime-assurance principle, while addressing a different object: deceptive thermal reports. The comparison also identifies missing deployment controls. An internal one-shot capability and payload hash do not authenticate a person or agent and do not implement RBAC, IAM, PKI, operating-system isolation, or a network policy-enforcement point.

Control, cost, and scalability

The action-dependent pilot improves realism over static acceptance counts by closing the actuator/plant loop and reporting IAE, SLA exposure, recovery, and fan effort. It also reveals an important negative boundary: common-mode bias delays recovery for both measurement-dependent controllers. The true-CRP benchmark shows sub-millisecond execution through $n_z = 100$ in the tested CPU environment. Theoretical arithmetic and storage remain quadratic per round, so larger zones should be partitioned or benchmarked on their target edge device. Fan-command effort is a control surrogate; without power instrumentation it cannot support an energy or consumption claim.

Role of the LLM

The architecture permits a language supervisor for semantic escalation and auditable proposals, but the available reactive and LLM archives are not a paired causal study. We therefore make no claim that either supervisor causes better thermal outcomes, that event triggering produces the historical latency reduction, or that an LLM adds measurable value in this task. The corrected deterministic runtime and periodic closed-loop pilot establish the mechanism independently of new LLM inference.

Smart-city relevance

The evaluated component operates inside one illustrative facility zone. Its relevance to smart cities is infrastructural: urban services depend on cloud and edge data centers, and future district-energy or city-level orchestration requires trustworthy facility state. The present study does not validate a city-scale deployment, cross-zone arbitration, or district-energy coupling.

9. Limitations

The evaluation is software based. The historical thermal series appears standardized, so its replay errors are dataset units rather than degrees Celsius. ENACT supplies workload/power context and CuCD-ID supplies attack timing; neither is interpreted as co-measured thermal telemetry. The new first-order plant uses declared illustrative parameters. It validates code paths and action-dependent behavior, not physical calibration, hardware latency, thermal safety, or transfer to another facility. A unified instrumented data-center or hardware-in-the-loop study is required for those claims.

The replay and closed-loop pilots use bounded designs. The ten-seed robustness extension stabilizes the estimation ranking, while control and LLM archives use three seeds and are descriptive. All 135 original closed-loop runs stayed below the SLA, so mitigation and unsafe-state recovery were uninformative; the 18-run hot-start appendix is an exploratory boundary, not a significance study. FPR–OWA+CRP missed many drift/freeze/replay runs, had 627 non-convergent ticks, and generated 39 fallbacks. A common bias delayed both measurement-dependent controllers. These adverse findings constrain any robustness claim.

The dynamic temporal score was replayed on fixed trajectories and executed actions. It diagnoses the estimator but cannot establish counterfactual closed-loop benefit. Fifty legacy estimates differ only at dominance gaps no larger than 10^{-12}; stable reporter IDs deterministically extend exact ties but do not guarantee bitwise cross-platform identity for near-equal floating-point values.

Security validation is process local. The runtime matrix covers malformed, stale, mutated, superseded, timeout, and authorization-reuse cases, but not host compromise, malicious code with direct actuator access, network attacks, identity lifecycle, authorization administration, or production isolation. The zero observed unsafe-execution count is not a formal proof. Energy, electrical consumption, and token use were not measured. Historical reactive and LLM paths are non-identical, no new LLM inference was run, and live asynchronous behavior remains untested.

10. Conclusions and Future Work

We presented and audited a two-path architecture that combines trust-graded FPR–OWA sensing, persistent CRP diagnostics, optional supervisory proposals, and deterministic command admission. The revised evidence supports three bounded conclusions. First, estimator accuracy is context dependent, while the persistent CRP contributes an integrated diagnostic signal. Second, the tested runtime rejects the observed unsafe and process-invalid cases and keeps command authority separate from probabilistic planning. Third, the illustrative closed loop exposes both useful tracking behavior and vulnerability to common-mode measurement bias; the true CRP remains inexpensive for the tested zone sizes.

The work therefore provides a reproducible software blueprint and a clearer experimental baseline, rather than a production-ready or hardware-validated safety system. Next steps are a unified instrumented-facility or hardware-in-the-loop campaign, identity and least-privilege enforcement around the process-local gate, larger and device-specific scalability tests, electrical power measurement, counterfactual evaluation of temporal models, and a genuinely paired reactive-versus-LLM study on a task where semantic reasoning may affect a measurable outcome.

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